BEHAVIOURAL BIOMETRICS

Signature & Handwriting Technology

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Signature and handwriting biometrics exploit the uniqueness of an individual's writing habits — shaped by years of neuromotor learning, personality, and physiology.

Modern digital systems automate what forensic document examiners have done manually for centuries.

Key Historical Milestones

1960s

First Automatic Signature Verification

Early electronic tablets capture pen position. Researchers at IBM publish first automated signature verification systems.

1977

Plamondon & Maarse Framework

Formal kinematic model for human handwriting — underpins dynamic feature theory still used today in modern systems.

1989

Hidden Markov Models Applied

HMM-based handwriting recognition developed; later forms the backbone of IBM ViaVoice and Google speech recognition.

2001

BioAPI Standard Established

ISO/IEC 19784 BioAPI provides programming interface for biometric systems including signature verification.

2014

Deep Learning Era Begins

LSTM and CNN approaches outperform classical methods; Google's Deep Mind handwriting synthesis demonstrates human-like generation.

2021+

Transformer-Based HTR

Vision Transformers (ViT) and TrOCR achieve near-human performance on HTR benchmarks like IAM and RIMES datasets.

- **Handwritten Text Recognition (HTR)** : is the task of automatically converting handwritten text in images into machine readable digital data
- **Vision Transformers (ViT):** is a deeplearning architecture designed for image understanding using Transformer model, originally developed for NLP.
- **Transformer based Optical Character Recognition(TrOCR)**: is a model developed by Microsoft.

Handwriting Dataset

- **IAM**: One of the most widely used English handwriting datasets
- **RIMES:** French handwritten document dataset

Technical Description: System Architecture Overview

- Signature & handwriting biometric systems follow a standard pipeline with two major branches based on input acquisition mode:

1. **ONLINE (DYNAMIC) SYSTEM**
2. **OFFLINE (STATIC) SYSTEM**

The **online system** captures *time-series behavioral data* in real time, while the **offline system** works from *static images*, making online systems generally more secure due to the richer dynamic information available.

ONLINE (DYNAMIC) SYSTEM

- **Acquisition** — A digitizer tablet/stylus or touchscreen captures X/Y coordinates, pressure, pen-tilt, and timestamps at 100–200Hz.
- **Pre-processing** — Noise filtering (Gaussian/Kalman), resampling to uniform time intervals, and duplicate point removal.
- **Segmentation** — Identifies pen-down/pen-up strokes and segments them into sub-units (allographs) using velocity minima.
- **Feature Extraction** — Computes dynamic features like velocity profile, acceleration, curvature, pressure, time series, and stroke count.
- **Classification** — DTW / HMM / SVM / LSTM compares the feature vector against enrolled templates.

CLASSIFICATION

- **DTW — Dynamic Time Warping**
  - A **temporal alignment algorithm** that compares two time-series sequences even if they differ in speed or timing
  - It "warps" the time axis to find the **optimal alignment** between the test signature and the template
  - Great for signatures because the same person may sign **faster or slower** each time
  - Measures the **minimum distance** between two sequences — lower distance = better match

- **HMM — Hidden Markov Model**
  - A **probabilistic/statistical model** that treats signature writing as a sequence of hidden states (e.g., strokes, curves)
  - Models the **temporal transitions** between writing states
  - Each enrolled signature trains an HMM; the test signature is scored against it using **likelihood probability**
  - Well-suited for capturing **sequential pen movement patterns**

- **SVM — Support Vector Machine**
  - A **supervised machine learning classifier** that finds the optimal boundary (hyperplane) separating genuine vs. forged signatures
  - Works in **high-dimensional feature space**
  - Very effective with **small training datasets** (few enrolled samples)
  - Outputs a binary decision: **genuine or forgery**

- **LSTM — Long Short-Term Memory**
  - A type of **deep learning / recurrent neural network (RNN)**
  - Specifically designed to learn **long-range dependencies** in sequential data
  - Learns complex temporal patterns in stroke sequences that simpler models may miss
  - Most powerful but requires **more training data**

Comparison

Method	Type	Strength	Weakness
DTW	Distance-based	Simple, interpretable	Sensitive to noise
HMM	Probabilistic	Good sequence modeling	Needs careful tuning
SVM	ML Classifier	Works with small data	Limited temporal modeling
LSTM	Deep Learning	Highest accuracy	Needs large dataset

In practice, **LSTM and SVM** tend to be most popular in modern systems, with LSTM dominating when sufficient training data is available.

OFFLINE (STATIC) SYSTEM

- **Acquisition** — A flatbed scanner (300–600 DPI), camera, or document digitizer captures a grayscale or binary image of the signature.
- **Pre-processing** — Binarization (Otsu/Niblack), noise removal, slant correction, and baseline normalization.
- **Segmentation** — Foreground/background separation, connected component analysis, stroke extraction, and zone identification.
- **Feature Extraction** — Global bounding box, aspect ratio, local pixel density, zoning, gradient histograms (HOG, LBP).
- **Classification** — SVM / CNN / Siamese network compares the extracted feature set against enrolled samples in a database.

Pre-processing Pipeline

- Pre-processing converts a raw scanned signature image into a **clean, standardized form** before feature extraction.

- Each step removes a specific type of distortion or variability.

  - **Binarization (Otsu/Niblack),**
  - **noise removal,**
  - **slant correction, and**
  - **baseline normalization**

1. Binarization

- Converts a **grayscale image into a black-and-white (binary) image** — signature pixels become black (1), background becomes white (0).
- Otsu's Method
  - Automatically finds the **optimal global threshold** value to separate foreground from background
  - Analyzes the image's **histogram** and picks the threshold that minimizes intra-class variance
  - Works well when lighting is **uniform** across the document
  - Fast and simple

  - **If pixel intensity > threshold → White (background)**
  - **If pixel intensity ≤ threshold → Black (signature)**

- Niblack's Method
  - Uses a **local/adaptive threshold** — calculates a different threshold for each pixel based on its neighborhood

Formula: T = mean + k × standard deviation

T = Threshold value (the decision boundary)
mean = Average pixel intensity in a local neighborhood
k = A constant (typically **−0.2**)
standard deviation = Spread of pixel intensities in that neighborhood

  - Better for **uneven lighting, shadows, or aged paper**
  - More computationally expensive but more accurate for difficult scans

2. Noise Removal

Eliminates **unwanted artifacts** introduced during scanning or from the paper itself.

Noise Type	Cause	Removal Method
Salt & pepper noise	Scanner sensor errors	Median filtering
Speckle noise	Paper texture	Gaussian smoothing
Small dots/blobs	Dust, paper grain	Morphological opening
Broken strokes	Low ink density	Morphological closing

3. Slant Correction

- Corrects the **angle/tilt** of the signature caused by how the person held the pen or placed the paper.

  - The same person may produce signatures **tilted left or right** across different instances
  - Slant makes feature comparison unreliable without correction.
  - Typical slant range corrected: **-30° to +30°**
  - Shear transformation is used specifically for **character-level italicness**

4. Baseline Normalization

- Standardizes the **position, size, and scale** of the signature so all signatures are comparable regardless of where on the page they were written.
- What it corrects: **Vertical position** — Shifts signature to a standard vertical baseline

Baseline = bottom-most consistent stroke line of the signature

- Size/Scale normalization:
  - Rescales signature to a **fixed bounding box** (e.g., 128×64 pixels)
  - Removes variability due to writing size differences
- Zone normalization:
  - Divides signature into **ascender / core / descender** zones
  - Aligns these zones consistently across samples

Offline System: Segmentation

Segmentation breaks the **clean binary image** (from pre-processing) into meaningful structural parts for feature extraction.

1. Foreground/Background Separation

   The first step — definitively deciding which pixels **belong to the signature** vs. the **paper/background**.

2. Connected Component Analysis (CCA)

   Identifies **distinct groups of connected pixels** — each group represents a separate ink element (letter, dot, stroke segment).

3. Stroke Extraction

   Extracts **individual pen strokes** — the fundamental drawing units of a signature.

4. Zone Identification

   Divides the signature into **spatial regions** to capture where different writing elements occur.

Offline System: Feature Extraction

1. Global Bounding Box
   - The **smallest rectangle** that completely encloses the entire signature.
2. Aspect Ratio
   - The **ratio of width to height** of the bounding box.
3. Local Pixel Density
   - Measures **how much ink exists in small local regions** across the signature.
4. Zoning
   - Divides the signature into **predefined spatial zones** and extracts features from each zone independently.
5. Gradient Histograms
   - Captures the **direction and strength of ink edges** — describing HOW strokes are oriented rather than just WHERE ink exists.

Offline System: Classification

Classification is the **final decision stage** — it takes the extracted feature vector and answers: **"Is this signature genuine or forged?“**

1. SVM — Support Vector Machine
   - A **supervised machine learning classifier** that finds the optimal mathematical boundary between genuine and forged signatures.
2. CNN — Convolutional Neural Network
   - A **deep learning model** that learns features DIRECTLY from raw signature images — no manual feature extraction needed.
3. Siamese Network
   - A **specialized deep learning architecture** designed specifically for **similarity comparison** — perfect for signature verification.

Handwriting Recognition (HTR) — Technical Description

Levels of Handwriting Recognition: **FOUR levels of HTR systems** depending on how much text is processed.

1. Isolated Character Recognition
   - Recognizes **single characters that are already separated**.
   - Each character is processed independently.
2. Word-Level Recognition
   - Recognizes **entire words as a unit** rather than individual characters.
3. Line-Level Recognition
   - Recognizes **entire lines of handwritten text**.
4. Page-Level Document HTR
   - Recognizes **complete handwritten pages**.

Standard HTR Processing Pipeline

FIVE major stages used to recognize handwriting.

- Step 1: Layout Analysis
  - **Understand the structure of the document**.
- Step 2: Line Segmentation
  - **Separate handwritten lines from the document**.
- Step 3: Feature Extraction
  - **Extract useful handwriting features from images**.
- Step 4: Sequence Modelling
  - **Model the sequence of characters in handwriting**.
- Step 5: Decoding + Language Model
  - **Convert probabilities into meaningful text output**.

Writer Identification vs. Signature Verification

Signature Verification (1:1)

Task:	Determine if query signature belongs to claimed identity
Input:	Probe signature + claimed identity (enrolled template reference)
Output:	Accept / Reject decision with confidence score
Threshold:	Single global or user-specific FAR-calibrated threshold
Models:	DTW, HMM, SVM, Siamese CNN with contrastive loss
Use Cases:	Bank cheque auth, e-document signing, access control

Writer Identification (1:N)

Task:	Identify the author of an unknown handwriting sample from a closed set
Input:	Unknown handwriting sample only (no claimed identity)
Output:	Ranked list of top-N most likely authors with confidence scores
Matching:	Exhaustive comparison against all N enrolled templates
Models:	SVM multi-class, kNN, ArcFace CNN, HMM likelihood ranking
Use Cases:	Forensic document analysis, historical archive attribution

Skilled Forgery Types: Random (no knowledge) → Simple (knows name) → Skilled (practiced copy) — each requires progressively harder discrimination.

Performance Metrics

FAR — False Accept Rate

~1–3%

For skilled forgeries (online)

FRR — False Reject Rate

~2–5%

Natural signer variability

EER — Equal Error Rate

~1–2%

Best online DTW/LSTM systems

WER — Word Error Rate

<5%

HTR on IAM dataset with LM